Leo J. Beck

Materials Scientist | Molecular Dynamics | Scientific ML

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Current Ph.D. student in Materials Science & Engineering at the University of Colorado Boulder, where I study computational methods to predict structural characteristics of hybrid organic inorganic perovskites (HOIPs) in the Heinz Research Group. My research connects local chemical environments to crystalline and macro-scale properties. I specialize in classical molecular dynamics and forcefield parameterization, as well as scientific machine learning. I also have hands-on experience with a broad range of complementary materials simulation techniques and experimental instrumentation.

My work combines molecular dynamics and machine learning to predict structure and properties of HOIPs. I develop Python-based scientific software to set up, run, process, and visualize large, multidimensional datasets, using Git/GitHub for version control and open-source contributions. I am seeking an internship to showcase my skills while gaining additional experience before my graduation in Spring 2028.

Education

2028 **Ph.D., Materials Science & Engineering** — University of Colorado Boulder, Boulder, CO.
Advisor: Hendrik Heinz.

2025 **M.S., Materials Science & Engineering** — The University of Colorado Boulder, Boulder, CO.
GPA: 3.90/4.00
Advisor: Hendrik Heinz.

2023 **B.S., Mathematics, B.S., Materials Science & Engineering**, — Clemson University, Clemson, SC.
GPA: 3.84/4.00

Research Experience

Oct 2026 – Aug 2023 | University of Colorado Boulder
Graduate Research Assistant | Heinz Group, Boulder, CO

- Calculated formation energy of Hybrid Organic Inorganic Perovskites (HOIPs) using LAMMPS.
- Developed Python software for structure creation, simulation setup, analysis, visualization, and modeling for high-throughput simulations; used Git/GitHub for version control and contributions.
- Built machine learning pipelines to predict structural properties of HOIPs that are central to material and device performance.
- Made a user-friendly GUI to effectively present scientific findings.
- Configured and executed Density Functional Theory (DFT) molecular optimization and dipole moment calculation of Self-Assembled Monolayers (SAMs) using ORCA.
- Collaborated with students, external research groups, and national laboratory scientists to translate ambiguous scientific questions into experiments, data workflows, and interpretable models.

molecular dynamics | machine learning | scientific software | Python | perovskite chemistry | GitHub

Summer 2025 | Helmholtz-Zentrum Berlin für Materialien und Energie (HZB)
Guest Researcher | Collaboration with FAIRmat / Draxl Group, Berlin, Germany

- Updated NOMAD LAMMPS parser in Python to handle a wide variety of input and output formats.
- Consulted on LAMMPS methods, options, and packages.
- Curated set of LAMMPS input and output files for future test cases.
- Participated in the FAIRmat workshop to share research, technical progress, and database strategies with an international team of software engineers and chemical researchers.

Python | data parsing | molecular dynamics | Github

Summer 2026 – Summer 2024 | Air Force Research Laboratory (AFRL)
High-Performance Computing Intern | Materials & Manufacturing Directorate, Fairborn, OH

- Performed replicated LAMMPS simulations to characterize shear strength and thermal conductivity as a function of surface environment within $Ti_3C_2T_x$ MXenes.
- Calculated slip probability vs. force and calculated logistic fits to extract inter-sheet shear strength.
- Automated structure creation, post-processing, and visualization in Python.
- Used NAMD to calculate absorption energy of dopamine (DAH+) on MXene surfaces with realistic pH and solvent.

molecular dynamics | Python | data analysis | Github | LAMMPS | NAMD | VMD

Publications

Published

1. I. Armstrong, J. Slocik, ***L. Beck***, D. Nepal, and C. Zhu, V. Varshney, H. Heinz, **Validated Reactive Force Field Quantifies MXene Interfacial Properties, Mechanics, and Thermal Transport**, *ACS Nano* (2026), 10.1021/acsnano.5c20046.
2. M. Karabin, I. Armstrong, ***L. Beck***, P. Apanel, M. Eisenbach, D. B. Mitzi, H. Terletska, H. Heinz, **Enhancing dimensionality prediction in hybrid metal halides via feature engineering and class-imbalance mitigation**, *Mach. Learn.: Sci. Technol.*, (2026) 10.1088/2632-2153/ae62cb.
3. B. Grant, T. Wanless, J. Crooks, ***L. Beck***, Y. Bandera, and S.H. Foulger. (2023). **Nonconformal electrochemical memristor through vapor phase polymerization of pyrrole**. 10.1021/acsaelm.3c00708.

Preprints, Submissions, & Manuscripts Under Review

4. Shuck, C. et al. (2025). **MXenes 2025: A Roadmap on Synthesis, Processing, and Applications**. 92–98. 10.26434/chemrxiv.15003918/v2

In Preparation

5. ***L. Beck***, M. Karabin, Y. Xie, Y. Jeong, J. Zhang, S. Barlow, S. Marder, D. B. Mitzi, H. Terletska, and H. Heinz, **Machine Learning Prediction of Hybrid Metal Halide Structure Families**, *planned for submission in December 2026*,
6. ***L. Beck***, S. Bekele, T. Muzzy, D. Nepal, H. Heinz, and V. Varshney, **Modeling of MXene Interfacial Thermal Conductivity & Shear Strength in poly-BZ via Atomistic Molecular Dynamics**, *planned for submission in March 2027*,

Recognition & Awards

Workplace & Academic

2023 — Clemson University: Gilbert Robinson Research Award
2019 — South Carolina High School State Geography Champion
2018 — Sonic Drive-In, Tega Cay SC: Employee of the Month

Selected GitHub Projects & Contributions

Active Projects

Organizer desktop app https://leojbeck.com/projects/organization-app/ 2026 – Present

Personal organizer app made for Windows desktop. Contains homepage, calendar, ToDos, projects, notes, and productivity. Productivity includes a mood map for tracking productivity, similar to Github contributions map.

Tauri | Docker | React

Open-Source Contributions

NOMAD - FAIRmat GitHub Summer 2025

Contributed LAMMPS parsing, test cases, and data to NOMAD.

Python | database | simulation parsing

Conferences, Talks, & Workshops

Poster Sessions

1. **Molecular Dynamics Simulation to Determine Dopamine Binding Mechanisms on MXene Surfaces**
   Poster, *American Physical Society (APS) 2026*, Denver, CO (2026)
   L. Beck, I. Armstrong, J. Slocik, D. Nepal, C. Zhu, V. Varshney, H. Heinz
2. **MXenes Under Stress Heat: Molecular Dynamics To Calculate Thermal And Mechanical Properties Of Mxene-Vitrimer Systems**
   Poster, *Air Force Research Laboratory Summer Poster Session*, Fairborn, OH, (2026)
   L. Beck, H. Heinz, D. Nepal, V. Varshney
3. **Paving the way for Future Electronics & Composites: Determining Shear Behavior of MXenes**
   Poster, *Air Force Research Laboratory Summer Poster Session*, Fairborn, OH, (2024)
   L. Beck, I. Armstrong, H. Heinz, D. Nepal, V. Varshney
4. **Molecular Dynamics Simulation and Machine Learning Prediction of Structures of Hybrid Organic Inorganic Perovskites**
   Poster, *American Chemical Society (ACS) 2024*, Denver, CO, (2024)
   L. Beck, I. Armstrong, H. Heinz
5. **Normal and Shear Force Controlled Reactor for Mechanochemistry**
   Poster, *Texas A&M University Summer Poster Session*, College Station, TX, (2022)
   L. Beck, E. Nwoye, J. Felts

Oral Presentations

1. **MD & ML Prediction of Hybrid Organic Inorganic Perovskites**
   Presentation, *Spring APS 2026*, Denver, CO, (2026)
   L. Beck, W. Weise, M. Karabin, I. Armstrong, W. Ashcraft, P. Apanel, H. Heinz

Workshops

1. **2025 HybriD3 Materials Theory Training Workshop**
   November 2025, Durham, NC | *Description*: Trained in atomistic simulation techniques, electronic structure theory, and FHIAims.

Skills

Materials Synthesis & Characterization

Working knowledge Thin-film device fabrication; photolithography; X-ray diffraction (XRD); spin coating; thermalgravimetric analysis (TGA); differential scanning calorimetry (DSC)

thin films | spin coating | device fabrication | thermal analysis

Programming & Computing

Expertise Python; NumPy; pandas; SciPy; Matplotlib; Bash; Git/GitHub; VS Code; Jupyter; Anaconda; Docker; Slurm; high-performance computing (HPC); Vim; LaTeX

Working knowledge MATLAB; C; C++; Julia, Intel Advisor, valgrind, Goose; FastAPI; React; TypeScript, VBA

Scientific software development for data analysis, visualization, and automation; design of graphical interfaces, command-line tools, and APIs; use of agentic AI to support programming and software development.

Python | Git/GitHub | scientific software | scientific computing | agentic AI | APIs | HPC

Scientific Analysis, Software, & Modeling

Expertise LAMMPS; NAMD; VMD; Packmol; Ovito; bash; reverse Monte Carlo; Markov chain Monte Carlo;

Working knowledge DFT with ORCA, VASP, and FHIAims; Finite Element Analysis (FEA); SolidWorks; ModeFrontier

Data reduction, visualization, fitting, and structural modeling of large, multidimensional datasets. Configured and executed DFT simulations of self-assembled monolayers (SAMs); analyzed molecular trajectories alongside experimental X-ray data.

molecular dynamics | DFT | Monte-Carlo simulations

Teaching Experience

Student Mentorship and Projects

1. **William Weise** — B.S. Student, Chemical Engineering
   Developed scripts for automatic structure generation; ran LAMMPS simulations of HOIPs.
2. **Naime Gokce** — B.S. Student, Biochemistry; Middle Tennessee State University AMPLIFY Undergraduate Researcher
   Performed forcefield parameter optimization for HOIPs.

Teaching Assistant

1. **University of Colorado Boulder, Physics 1140 — Experimental Physics (Spring 2026)**
   Taught introductory physics laboratory sections.
2. **University of Colorado Boulder, Physics 2220 — Classical Mechanics (Spring 2026)**
   Graded upper-division physics assignments; proctored exams.
3. **University of Colorado Boulder, Physics 1110 - Physics with Calculus 1 (Fall 2023)**
   Led 4 recitations to a total of 90 students; graded assignments; proctored exams.